



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Vectors

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- * Representing Vectors as Diagrams
- * Magnitude of a Vector
- * Position & Displacement Vectors
- * Finding Vector Paths
- * Problem Solving with Vectors



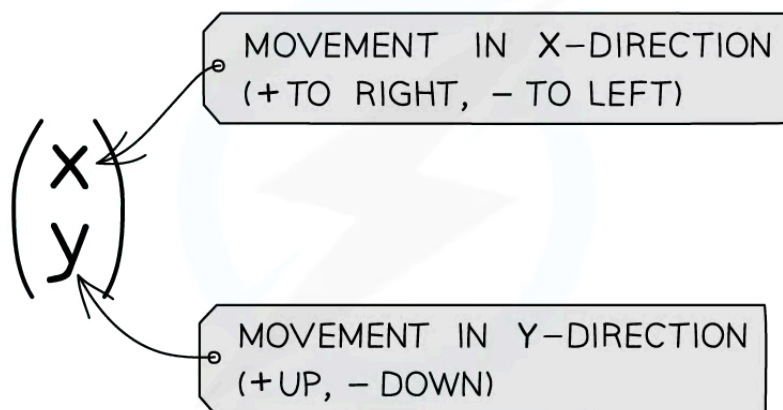
Your notes

Introduction to Column Vectors

Basic Vectors

What are column vectors?

- A **column vector** can be used to **describe** how to get **from one point to another point**
 - This is also called a **translation vector**
- $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$ means **6 units to the right** and **3 units up**



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How do I add and subtract column vectors?

- **Adding and subtracting vectors** is done by looking at the **top** numbers and **bottom** numbers **separately**
- To **add** column vectors
 - Add the top numbers together
 - Add the bottom numbers together



Your notes

$$\bullet \begin{pmatrix} 5 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 5+3 \\ 2+(-1) \end{pmatrix} = \begin{pmatrix} 8 \\ 1 \end{pmatrix}$$

- To **subtract** column vectors

- Subtract the second top number from the first

- Subtract the second bottom number from the first

$$\bullet \begin{pmatrix} 5 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 5-3 \\ 2-(-1) \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

How do I multiply a vector by a scalar?

- A **scalar** is **number** not a **vector**

- It does not have a direction

- To **multiply** a column vector by a **scalar**

- Multiply the top number by the scalar

- Multiply the bottom number by the scalar

$$\bullet 3 \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \times 2 \\ 3 \times (-1) \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix}$$

How do I write an expression as a single column vector?

- You need to follow the **order of operations**

$$\bullet 2 \begin{pmatrix} 5 \\ 2 \end{pmatrix} + 5 \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

- **STEP 1**

Multiply each vector by the scalar in front of it

$$\bullet \begin{pmatrix} 2 \times 5 \\ 2 \times 2 \end{pmatrix} + \begin{pmatrix} 5 \times 3 \\ 5 \times (-1) \end{pmatrix} = \begin{pmatrix} 10 \\ 4 \end{pmatrix} + \begin{pmatrix} 15 \\ -5 \end{pmatrix}$$

- **STEP 2**

Add or subtract the new column vectors

$$\bullet \begin{pmatrix} 10+15 \\ 4+(-5) \end{pmatrix} = \begin{pmatrix} 25 \\ -1 \end{pmatrix}$$



Your notes

**Worked Example**

$$\mathbf{a} = \begin{pmatrix} p \\ 3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

Given that $2\mathbf{a} + 3\mathbf{b} = \begin{pmatrix} 4 \\ q \end{pmatrix}$, find the value of p and the value of q .

Write the left-side side as one vector

Multiple each vector by the scalar in front of it

$$\begin{pmatrix} 2p \\ 6 \end{pmatrix} + \begin{pmatrix} -6 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ q \end{pmatrix}$$

Add the vectors together

$$\begin{pmatrix} 2p - 6 \\ 9 \end{pmatrix} = \begin{pmatrix} 4 \\ q \end{pmatrix}$$

The top components are equal

Form and solve an equation

$$2p - 6 = 4$$

$$2p = 10$$

$$p = 5$$

The bottom components are equal

$$9 = q$$

$$\mathbf{p} = 5 \text{ and } \mathbf{q} = 9$$



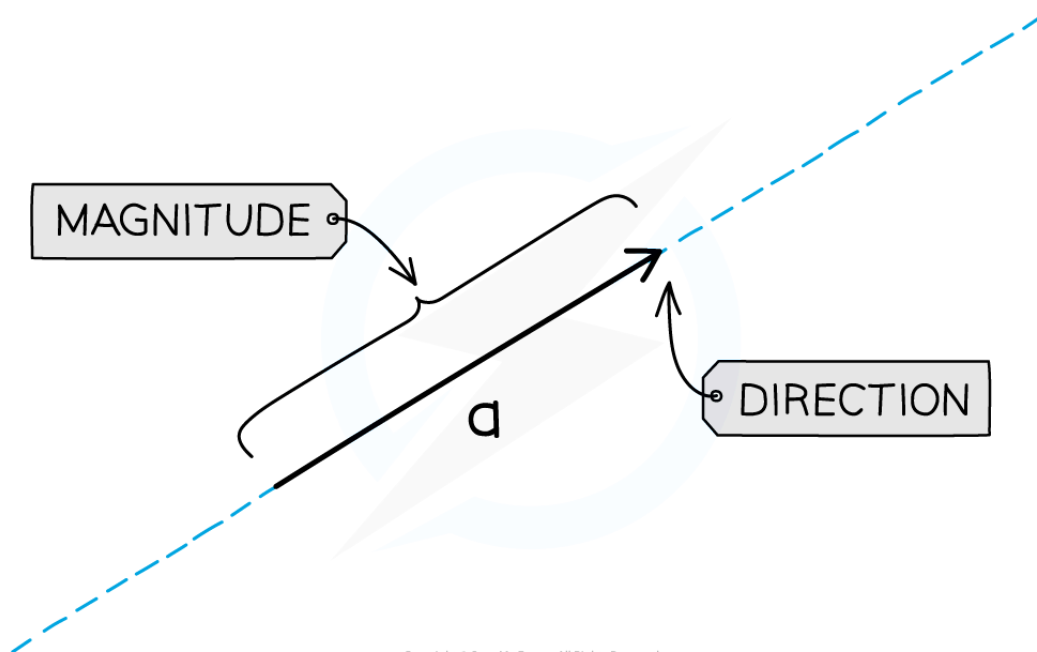
Your notes

Representing Vectors as Diagrams

Vector Diagrams

How can I represent a vector visually?

- A **vector** has **both a size (magnitude) and a direction**
 - You need to draw a **line** to show the **size of the vector**
 - You also need to draw an **arrow** to show the **direction of the vector**



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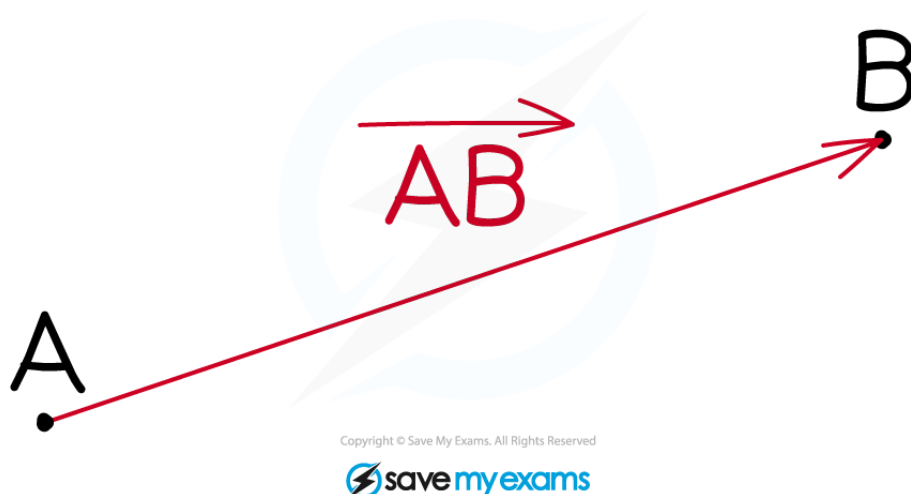


- Vectors are written in **bold** when typed to **show** that they are a vector and not a scalar
 - When writing a vector in an exam you should **underline** the letter to show it is a vector
 - **a** when **typed** and **a** when **handwritten**
 - You will **not lose marks** if you forget to underline vectors



Your notes

- If a vector **starts at A** and **ends at B** we can write it as $\vec{AB}$
 - Here the arrow will **point toward B**
 - Vector $\vec{BA}$ will have the **same length** but **point toward A**



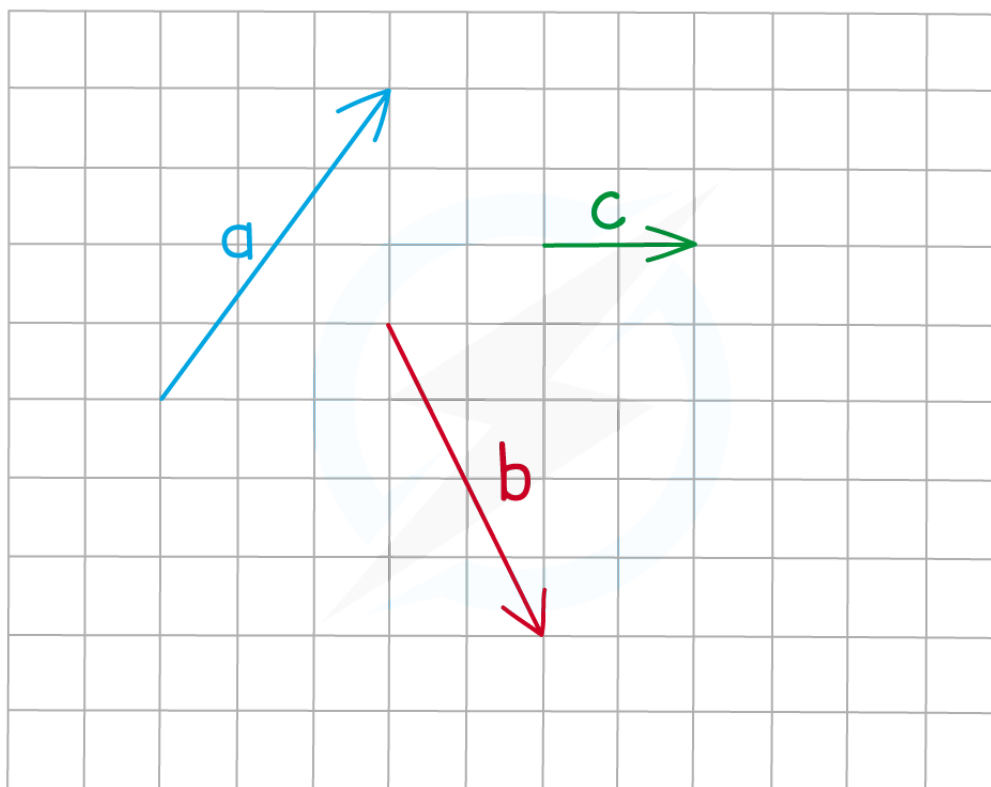
How do I draw a vector on a grid?

- You can draw a vector **anywhere** on a grid
 - Just make sure it has the **correct length** and the **correct direction**
- To draw the vector $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$
 - **Pick** a point on the grid and draw a **dot** there
 - Count 3 units to the right and 4 units up and **draw another dot**
 - Draw a **line between the two dots**
 - Put an **arrow** on the line **pointing toward the second dot**
- Look out for **negatives** and **zeroes**
 - $\mathbf{b} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$ goes **2 to the right** and **4 down**



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- $\mathbf{c} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ goes 2 to the right but does not go up or down



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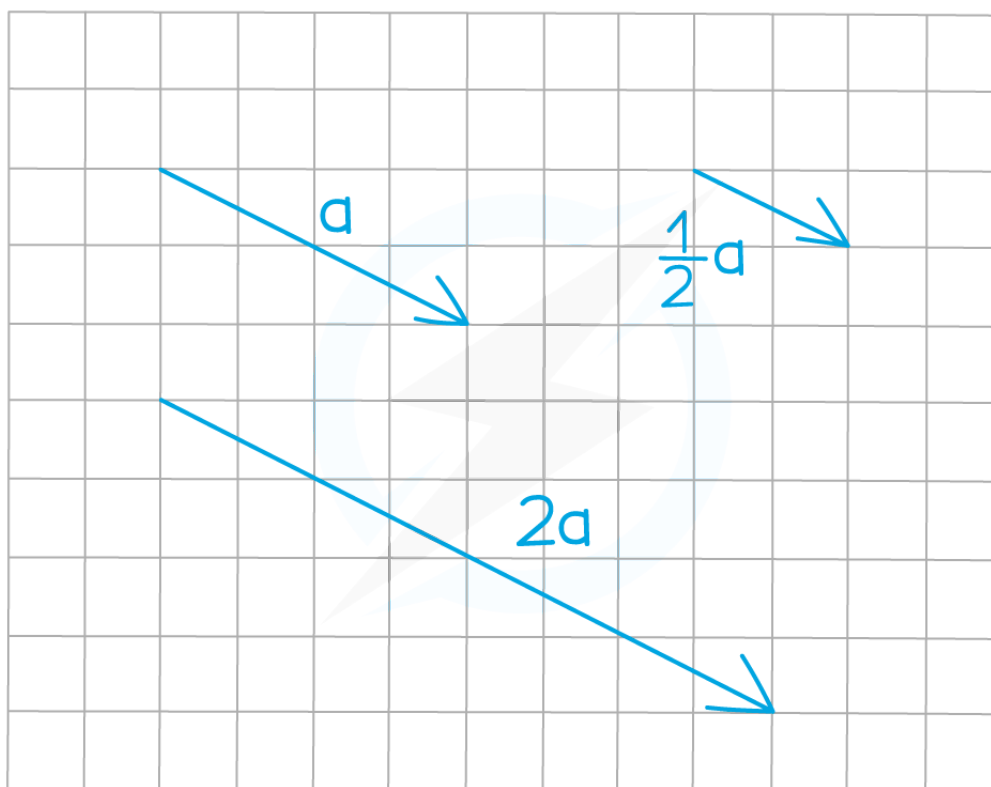
What happens when I multiply a vector by a scalar?

- When you **multiply** a vector by a **positive scalar**:
 - The **direction** stays the **same**
 - The **length** of the vector is **multiplied by the scalar**
- For example, $\mathbf{a} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$



Your notes

- $2\mathbf{a} = \begin{pmatrix} 8 \\ -4 \end{pmatrix}$ will have the **same direction** but **double the length**
- $\frac{1}{2}\mathbf{a} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ will have the **same direction** but **half the length**



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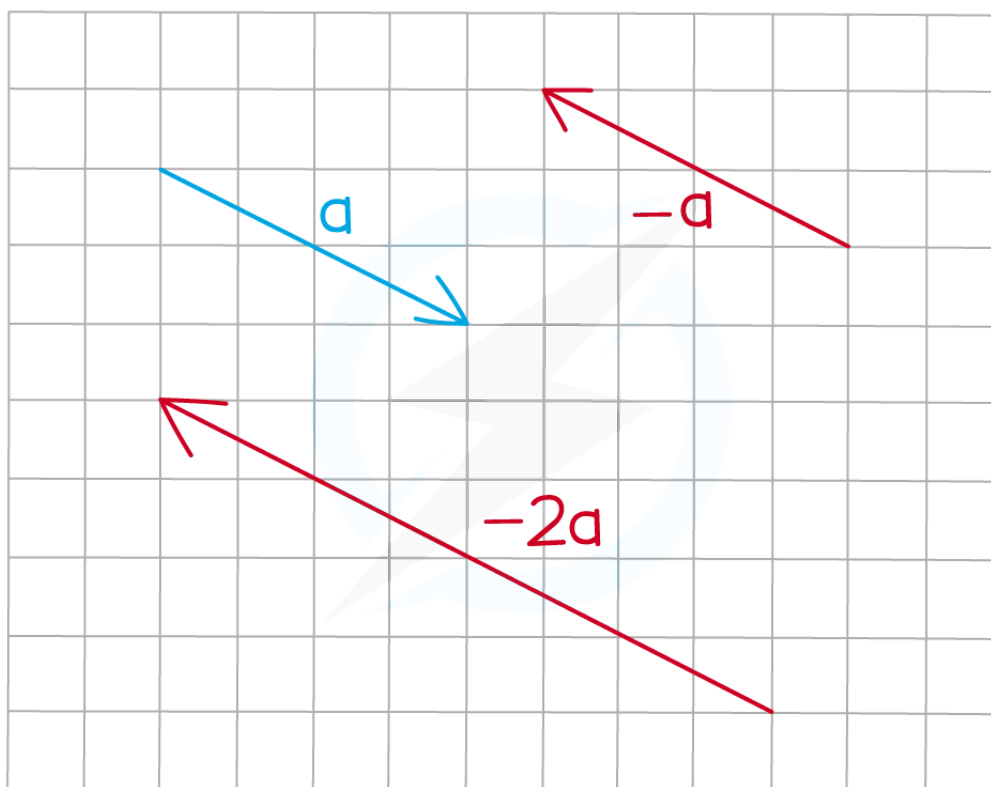


- When you **multiply** a vector by a **negative scalar**:
 - The **direction** is **reversed**
 - The **length** of the vector is **multiplied** by the **number after the negative sign**
- For example, $\mathbf{a} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$



Your notes

- $-\mathbf{a} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$ will be in the **opposite direction** and its **length will be the same**
- $-2\mathbf{a} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}$ will be in the **opposite direction** and its **length will be doubled**



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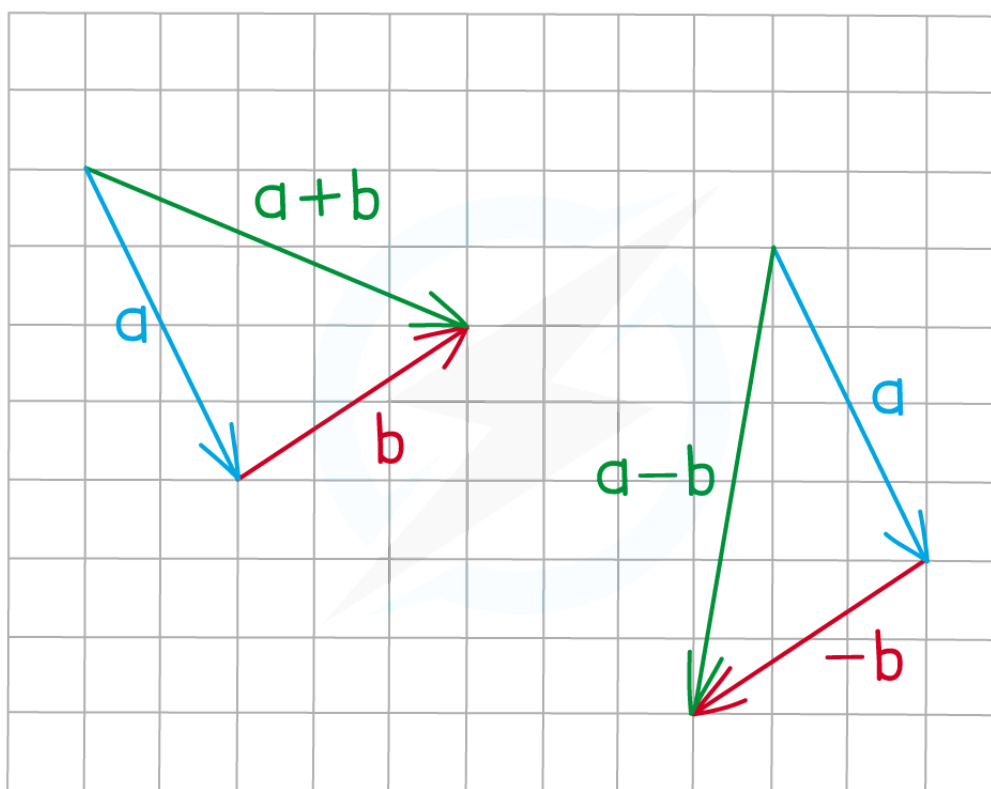
What happens when I add or subtract vectors?

- To **draw** the vector $\mathbf{a} + \mathbf{b}$
 - Draw the vector $\mathbf{a}$
 - Draw the vector $\mathbf{b}$ starting at the endpoint of $\mathbf{a}$



Your notes

- Draw a line that **starts at the start of $\mathbf{a}$** and **ends at the end of $\mathbf{b}$**
- To **draw the vector $\mathbf{a} - \mathbf{b}$**
 - Draw the vector $\mathbf{a}$
 - Draw the vector $-\mathbf{b}$ starting at the endpoint of $\mathbf{a}$
 - Draw a line that **starts at the start of $\mathbf{a}$** and **ends at the end of $-\mathbf{b}$**



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Worked Example

The points A, B and C are shown on the following coordinate grid.

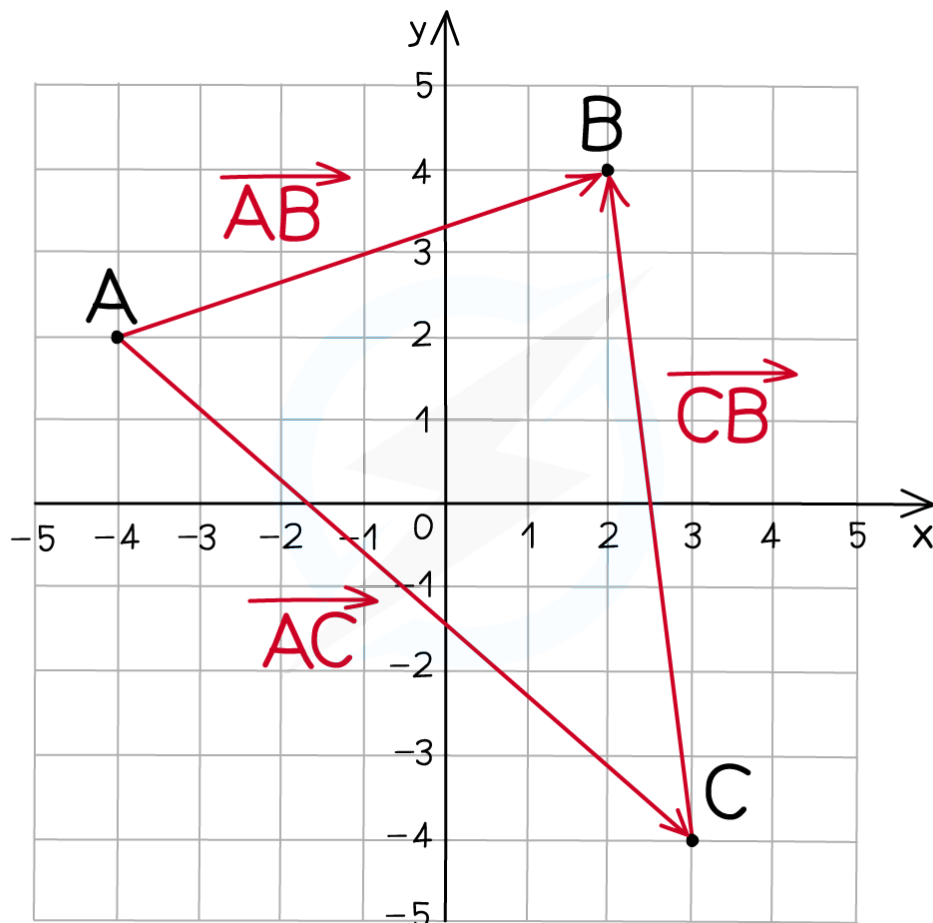


Your notes

(a)

Write the vectors $\vec{AB}$, $\vec{AC}$ and $\vec{CB}$ as column vectors.

Start by drawing the three vectors onto the grid



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From A to B, it is 6 to the right and 2 up

$$\vec{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$$

From A to C, it is 7 to the right and 6 down



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$$\overrightarrow{AC} = \begin{pmatrix} 7 \\ -6 \end{pmatrix}$$

From C to B, it is 1 to the left and 8 up

$$\overrightarrow{CB} = \begin{pmatrix} -1 \\ 8 \end{pmatrix}$$

(b)

Without using any calculations, explain why $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

The vector goes from A to B, then from B to C, then from C back to A

The vector returns to its starting point



Your notes

Magnitude of a Vector

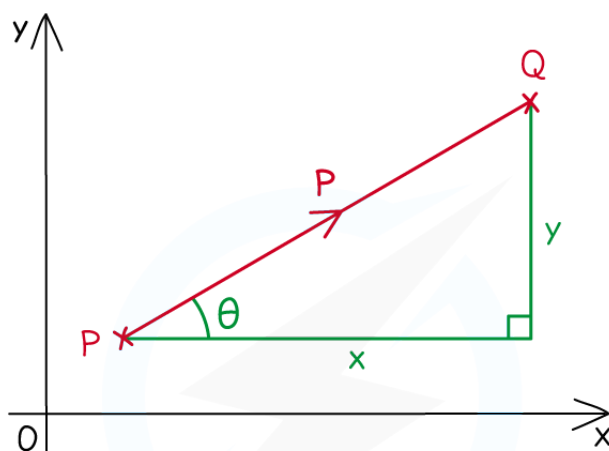
Magnitude of a Vector

How do I find the magnitude of a vector?

- The **magnitude** of a **vector** is its **length** (distance)
 - It is also called the **modulus**
 - This is always a **positive** value
 - The direction of the vector is irrelevant
- The magnitude of $\vec{AB}$ is written $|\vec{AB}|$
 - The magnitude of $\mathbf{a}$ is written $|\mathbf{a}|$
- Depending on the use of the vector, the magnitude of a vector represents **different quantities**
 - For velocity, magnitude would be speed
 - For a force, magnitude would be the strength of the force (in Newtons)
- In **component** form, the magnitude is the **hypotenuse** of a right-angled triangle
 - Use **Pythagoras' theorem** to find the magnitude
 - The magnitude of $\mathbf{a} = \begin{pmatrix} x \\ y \end{pmatrix}$
 - $|\mathbf{a}| = \sqrt{x^2 + y^2}$



Your notes



THE DISTANCE PQ IS CALLED THE
MAGNITUDE (OR MODULUS) OF
THE VECTOR P

THIS IS DENOTED BY $|P|$ OR $|\vec{PQ}|$

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Examiner Tips and Tricks

- If there is no diagram, sketch one!
- You can sketch a vector and use it to form a right-angled triangle



Worked Example

Consider two points $A(-3, 5)$ and $B(7, 1)$.

(a) Write down the column vector $\vec{AB}$.



Your notes

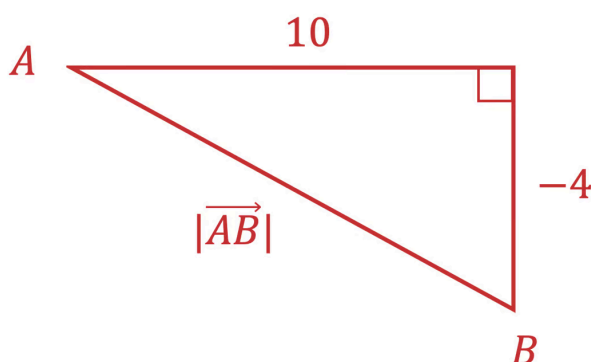
Find the horizontal and vertical distances between the two points
Subtract the x and y components of A from B

$$\vec{AB} = \begin{pmatrix} 7 - -3 \\ 1 - 5 \end{pmatrix}$$

$$\vec{AB} = \begin{pmatrix} 10 \\ -4 \end{pmatrix}$$

(b) Find the modulus of vector $\vec{AB}$.

Sketching a diagram of the vector $\vec{AB}$ can help



Apply Pythagoras' theorem to the x and y components of $\vec{AB}$

$$\begin{aligned} |\vec{AB}| &= \sqrt{10^2 + (-4)^2} \\ &= \sqrt{100 + 16} \\ &= \sqrt{116} \end{aligned}$$

$$|\vec{AB}| = 2\sqrt{29}$$

(c) Briefly explain why $|\vec{BA}| = |\vec{AB}|$.

The magnitude of a vector is it's 'size'

Direction of the vector is ignored



Your notes

 $|\vec{BA}| = |\vec{AB}|$ since both vectors have the same distance

Another vector, $\vec{CD}$, has three times the magnitude of vector $\vec{AB}$.

(d) Write down a possible column vector for $\vec{CD}$.

Being three times $|\vec{AB}|$ means the vector $\vec{AB}$ is three times longer

One way to find a vector is to multiply each component of the vector $\vec{AB}$ by 3 or -3

$$\vec{CD} = 3\vec{AB} = 3 \begin{pmatrix} 10 \\ -4 \end{pmatrix}$$

$$\vec{CD} = \begin{pmatrix} 30 \\ -12 \end{pmatrix}$$

Another possible answer is $\vec{CD} = \begin{pmatrix} -30 \\ 12 \end{pmatrix}$



Your notes

Position & Displacement Vectors

Position & Displacement Vectors

What are position vectors?

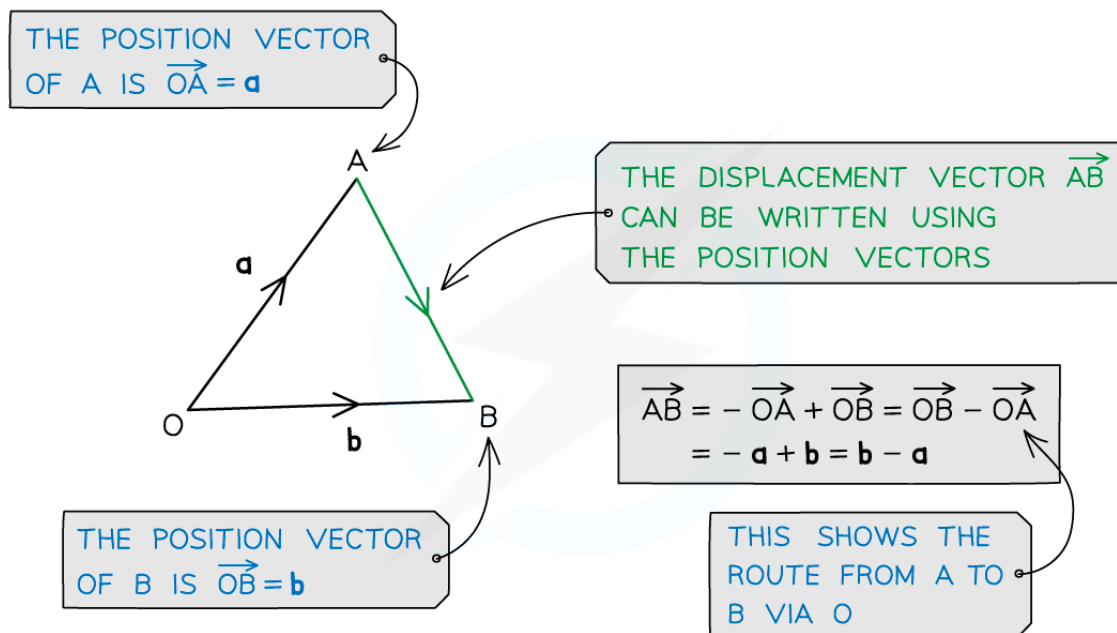
- A **position vector** describes where a specific **point**, A , is, relative to a fixed **origin**, O
 - **Lower-case bold** (or underlined) letters are used
 - The point A has position vector $\mathbf{a} = \overrightarrow{OA}$
- Their **components** are equal to their **coordinates**
 - The point with coordinates $(3, -2)$ has position vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ from the origin

What are displacement vectors?

- A **displacement vector** describes the **direction and distance** between **two points**
 - The displacement vector from A to B is $\overrightarrow{AB}$
 - How to get from A to B
- If the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ relative to O
 - then A to B is the same as A to O ($-\mathbf{a}$) followed by O to B ($\mathbf{b}$)
 - $\overrightarrow{AB} = -\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$
 - This is a useful rule to remember



Your notes



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Examiner Tips and Tricks

- You may need to draw an origin, O, on to a diagram to be able to sketch position vectors.



Worked Example

The points P and Q have position vectors $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ -10 \end{pmatrix}$ respectively.

Find and simplify the vector $\vec{PQ}$.

Let $\mathbf{p}$ and $\mathbf{q}$ be position vectors of P and Q

$\vec{PQ}$ is the displacement vector from P to Q

Use the rule that $\vec{AB} = -\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$



Your notes

$$\vec{PQ} = \mathbf{q} - \mathbf{p}$$

Substitute in $\mathbf{p}$ and $\mathbf{q}$

$$\vec{PQ} = \begin{pmatrix} 6 \\ -10 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Expand and simplify

$$\begin{aligned} &= \begin{pmatrix} 6 - 3 \\ -10 - 2 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ -12 \end{pmatrix} \end{aligned}$$

$$\vec{PQ} = \begin{pmatrix} 3 \\ -12 \end{pmatrix}$$

You can also get this answer by seeing what vector must be added to $\mathbf{p}$ to get $\mathbf{q}$



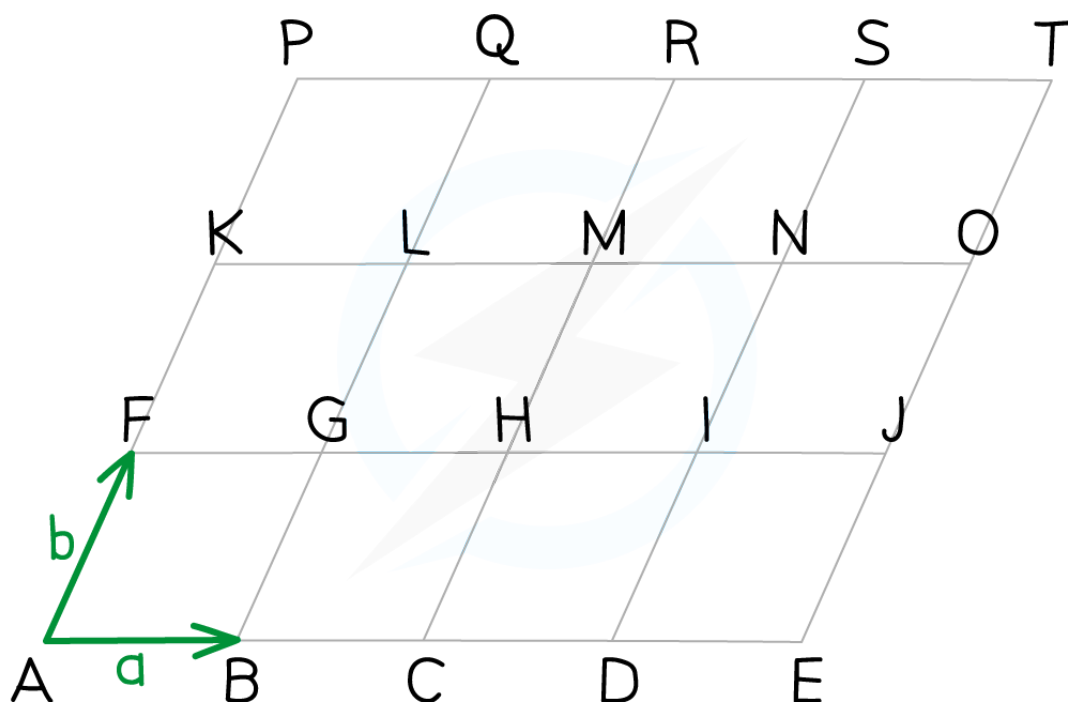
Your notes

Finding Vector Paths

Finding Vector Paths

How do I find the vector between two points?

- A **vector path** is a path of vectors taking you from a **start point** to an **end point**
- The following grid is made up entirely of parallelograms
 - The vectors **a** and **b** defined as marked in the diagram:
 - Any vector that goes **horizontally to the right** along a side of a parallelogram will be equal to **a**
 - Any vector that goes **up diagonally to the right** along a side of a parallelogram will be equal to **b**



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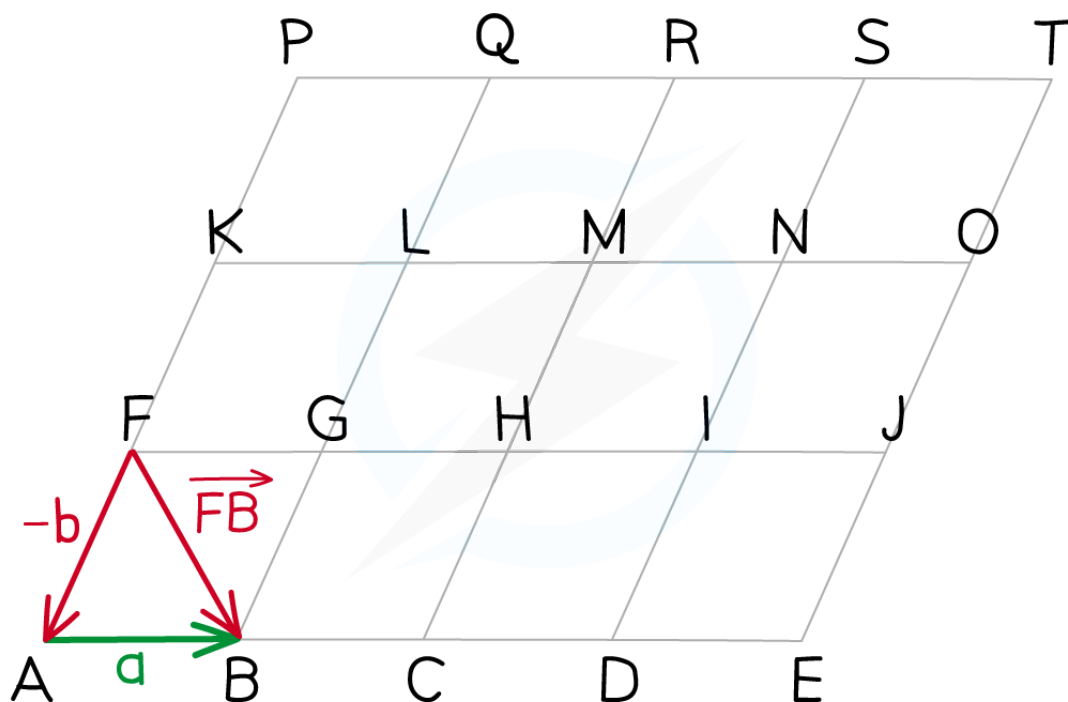


- To find the vector between two points



Your notes

- Count how many times you need to go **horizontally to the right**
 - This will tell you how many **a**'s are in your answer
- Count how many times you need to go up **diagonally to the right**
 - This will tell you how many **b**'s are in your answer
- Add the **a**'s and **b**'s together
 - E.g. $\vec{AR} = 2\mathbf{a} + 3\mathbf{b}$
- You will have to put a **negative** in front of the vector if it goes in the **opposite direction**
 - $-\mathbf{a}$ is one length **horizontally to the left**
 - $-\mathbf{b}$ is one length **down diagonally to the left**
 - E.g. $\vec{FB} = -\mathbf{b} + \mathbf{a}$ or $\vec{FB} = \mathbf{a} - \mathbf{b}$
 - Likewise, $\vec{BF} = -\vec{FB} = -(-\mathbf{b} + \mathbf{a}) = \mathbf{b} - \mathbf{a}$



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Your notes

- It is possible to describe **any vector** that goes from one point to another in the above diagram in terms of **a** and **b**



Examiner Tips and Tricks

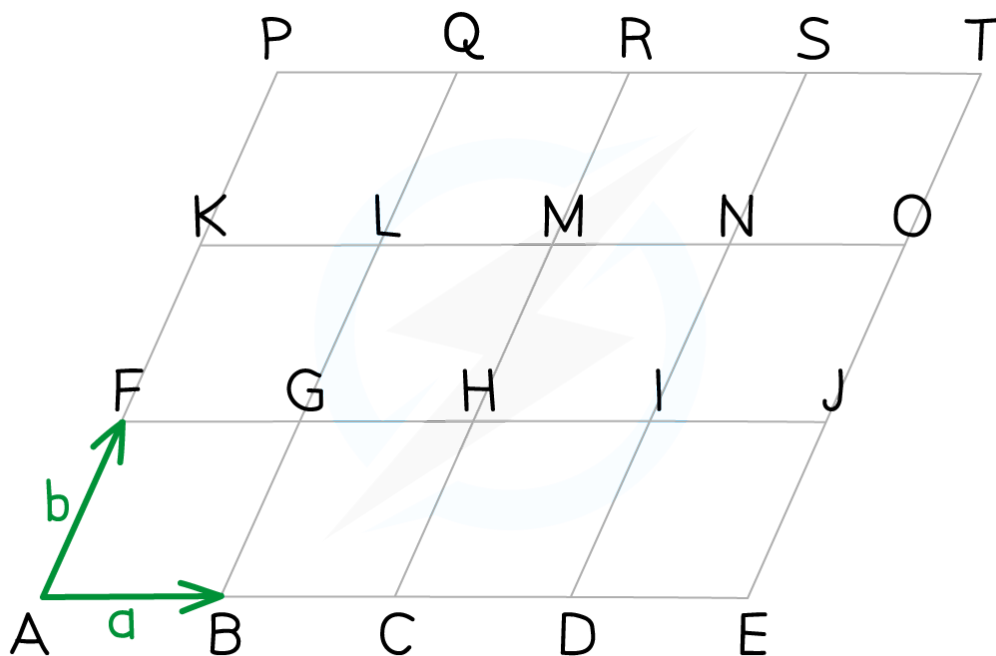
- Mark schemes will accept different correct paths, as long as the final answer is fully simplified
- Check for symmetries in the diagram to see if the vectors given can be used anywhere else



Worked Example

The following diagram consists of a grid of identical parallelograms.

Vectors **a** and **b** are defined by $\mathbf{a} = \overrightarrow{AB}$ and $\mathbf{b} = \overrightarrow{AF}$.



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Write the following vectors in terms of **a** and **b**.



Your notes

a) $\overrightarrow{AE}$

To get from A to E we need to follow vector **a** four times to the right

$$\begin{aligned}\overrightarrow{AE} &= \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} \\ &= \mathbf{a} + \mathbf{a} + \mathbf{a} + \mathbf{a}\end{aligned}$$

$$\overrightarrow{AE} = 4\mathbf{a}$$

b) $\overrightarrow{GT}$

There are many ways to get from G to T

One option is to go from G to Q (**b** twice), and then from Q to T (**a** three times)

$$\begin{aligned}\overrightarrow{GT} &= \overrightarrow{GL} + \overrightarrow{LQ} + \overrightarrow{QR} + \overrightarrow{RS} + \overrightarrow{ST} \\ &= \mathbf{b} + \mathbf{b} + \mathbf{a} + \mathbf{a} + \mathbf{a}\end{aligned}$$

$$\overrightarrow{GT} = 3\mathbf{a} + 2\mathbf{b}$$

c) $\overrightarrow{EK}$

There are many ways to get from E to K

One option is to go from E to O (**b** twice), and then from O to K (**-a** four times)

$$\begin{aligned}\overrightarrow{EK} &= \overrightarrow{EJ} + \overrightarrow{JO} + \overrightarrow{ON} + \overrightarrow{NM} + \overrightarrow{ML} + \overrightarrow{LK} \\ &= \mathbf{b} + \mathbf{b} - \mathbf{a} - \mathbf{a} - \mathbf{a} - \mathbf{a}\end{aligned}$$

$$\overrightarrow{EK} = 2\mathbf{b} - 4\mathbf{a}$$

$-4\mathbf{a} + 2\mathbf{b}$ is also acceptable



Your notes

Problem Solving with Vectors

Vector Problem Solving

What are vector proofs?

- Vectors can be used to **prove** things that are true in geometrical diagrams
 - **Vector proofs** can be used to find additional information that can help us to **solve problems**

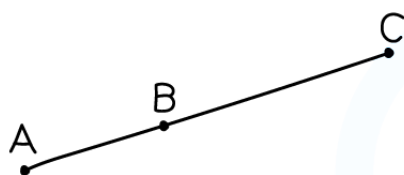
How do I know if two vectors are parallel?

- Two vectors are **parallel** if one is a **scalar multiple** of the other
 - This means if **b** is parallel to **a**, then **b = ka**
 - where **k** is a **constant number** (scalar)
- For example, $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$
 - $\begin{pmatrix} 2 \\ 6 \end{pmatrix} = 2 \times \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ so **b = 2a**
 - **b** is a scalar multiple of **a**, so **b** is **parallel** to **a**
- If the scalar multiple is **negative**, then the vectors are **parallel** and in **opposite** directions
 - $\mathbf{c} = \begin{pmatrix} -3 \\ -9 \end{pmatrix} = -3\mathbf{a}$
 - **c** is **parallel** to **a** and in the **opposite** direction
- If two vectors **factorise** with a **common bracket**, then they are parallel
 - They can be written as **scalar multiples**
- For example
 - $9\mathbf{a} + 6\mathbf{b}$ factorises to $3(3\mathbf{a} + 2\mathbf{b})$
 - $12\mathbf{a} + 8\mathbf{b}$ factorises to $4(3\mathbf{a} + 2\mathbf{b})$
 - This means $12\mathbf{a} + 8\mathbf{b} = \frac{4}{3}(9\mathbf{a} + 6\mathbf{b})$

- so they are **scalar multiples** of each other
- and therefore **parallel**

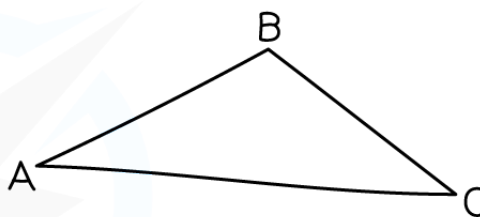
How do I know if three points lie on a straight line?

- You may be asked to prove that three points lie on a straight line
 - Points that lie on a straight line are **collinear**
- To show that the points A , B and C are **collinear**
 - prove that two line segments are **parallel**
 - and show that there is at least **one point** that **lies on both** segments
 - This makes them parallel and **connected** (not parallel and side-by-side)
- For example, if you show that $\vec{BC} = 2\vec{AB}$ then
 - the line segments AB and BC are parallel
 - and they have a **common point**, B
 - So A , B and C must be collinear
- Similarly, $\vec{AC} = 3\vec{AB}$ means AC and AB are parallel
 - and they have a common point, A
 - so A , B and C must be collinear



A, B, C COLLINEAR

$\vec{AB}$ PARALLEL TO $\vec{AC}$ AND $\vec{BC}$



A, B, C NOT COLLINEAR

$\vec{AB}$ NOT PARALLEL TO $\vec{AC}$ OR $\vec{BC}$

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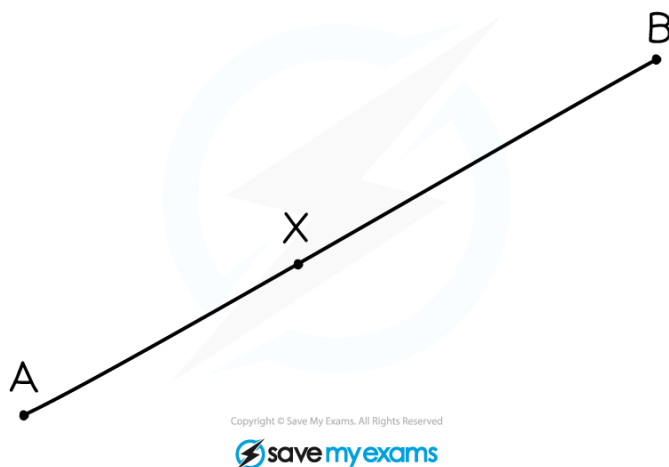


Your notes

How do I use ratios in vector paths?



Your notes



- Convert **ratios** into **fractions**
- In the example shown, if $AX : XB = 3 : 5$ then

- $\vec{AX} = \frac{3}{8} \vec{AB}$

- $\vec{XB} = \frac{5}{8} \vec{AB}$

- The ratio 3:5 has $3 + 5 = 8$ parts

- Always **check** which ratio you are being asked for

- $\vec{AX} = \frac{3}{5} \vec{XB}$

- $\vec{XB} = \frac{5}{3} \vec{AX}$



Worked Example

The diagram shows trapezium OABC.

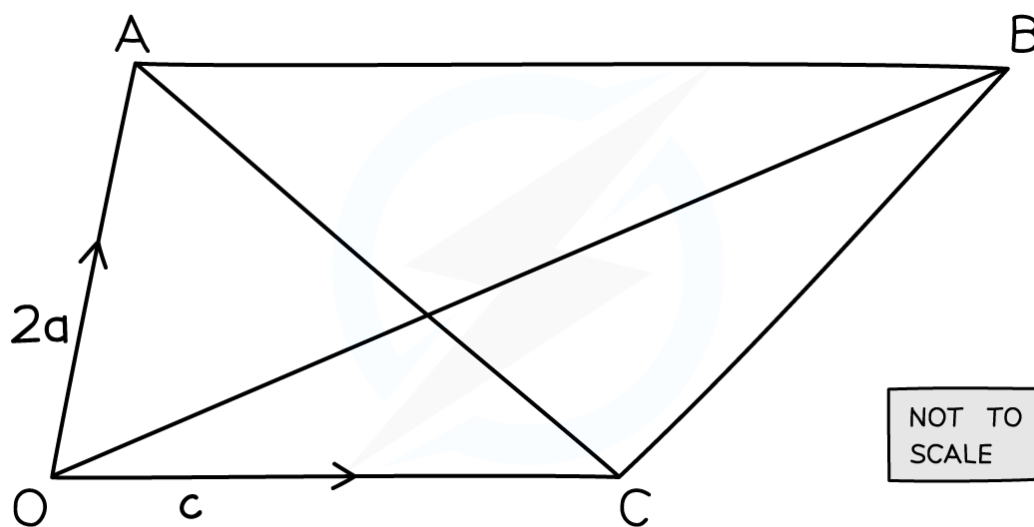


Your notes

$$\vec{OA} = 2\mathbf{a}$$

$$\vec{OC} = \mathbf{c}$$

AB is parallel to OC, with $\vec{AB} = 3\vec{OC}$.



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(a) Find expressions for vectors $\vec{OB}$ and $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.

$$\vec{AB} = 3\vec{OC} \text{ and } \vec{OC} = \mathbf{c} \text{ so } \vec{AB} = 3\mathbf{c}.$$

$$\begin{aligned}\vec{OB} &= \vec{OA} + \vec{AB} \\ &= 2\mathbf{a} + 3\mathbf{c}\end{aligned}$$

$$\vec{OB} = 2\mathbf{a} + 3\mathbf{c}$$

$$\begin{aligned}\vec{AC} &= \vec{AO} + \vec{OC} \\ &= -\vec{OA} + \vec{OC} \\ &= -2\mathbf{a} + \mathbf{c}\end{aligned}$$

$$\vec{AC} = \mathbf{c} - 2\mathbf{a}$$



Your notes

(b) Point P lies on AC such that $AP : PC = 3 : 1$.

Find expressions for vectors $\vec{AP}$ and $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.

$$AP : PC = 3 : 1 \text{ means that } \vec{AP} = \frac{3}{3+1} \vec{AC} = \frac{3}{4} \vec{AC}$$

$$\vec{AP} = \frac{3}{4} \vec{AC} = \frac{3}{4} (-2\mathbf{a} + \mathbf{c})$$

$$\vec{AP} = -\frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{c}$$

$$\begin{aligned} \vec{OP} &= \vec{OA} + \vec{AP} \\ &= 2\mathbf{a} + \left(-\frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{c}\right) \end{aligned}$$

$$\vec{OP} = \frac{1}{2}\mathbf{a} + \frac{3}{4}\mathbf{c}$$

(c) Hence, prove that point P lies on line OB , and determine the ratio $\vec{OP} : \vec{PB}$.

To show that O , P , and B are collinear (lie on the same line), note that

$$\vec{OP} = \frac{1}{2}\mathbf{a} + \frac{3}{4}\mathbf{c} = \frac{1}{4}(2\mathbf{a} + 3\mathbf{c})$$

$$\vec{OP} = \frac{1}{4}(2\mathbf{a} + 3\mathbf{c}) = \frac{1}{4}\vec{OB}$$

$$\vec{OP} = \frac{1}{4}\vec{OB} \text{ therefore } OP \text{ is parallel to } OB$$

and so P must lie on the line OB

$$\text{If } \vec{OP} = \frac{1}{4}\vec{OB} \text{ then } \vec{PB} = \frac{3}{4}\vec{OB}$$

$$\vec{OP} : \vec{PB} = 1 : 3$$